ELEMENT A

PROBLEM EVALUATION & ELABORATION

The Engineering Tech 1 students were given two problems to solve. Either the problem with the chord being too tall or AC dripping. The class took into consideration which problem would suit them better to solve for them. Over half a year since the class had started, the class experienced many problems to solve and was taught how to identify the problem, plan it, solve it, iterate on it, and reflect on it. The class was taught how to solve problems, work on Onshape, and make several sketches of the solutions they planned. The class is meant to improve the students' thinking skills and problem solving skills to understand how engineering works.

During our observation on how these two problems have worked over half a year, many of us decided to pick the chord problem. However, some of us also picked a solution for the condensation problem with the AC. Through our observations over the 6 months, the AC seems to drip a lot, especially on humid days, causing the AC to drip more than it would on a normal day. The AC is often seen dripping onto the floor and causing the floor to have erosion and water on the ground.

The problem we observed at MHS is just outside of the classroom or any AC that is on that building and is pointing outwards to the path where students walk on a daily basis. This problem also worsens as the day goes on due to the temperature rising and causing more condensation to happen. The AC either needs a drip pan or a tube that could redirect or collect the water that the AC is dripping. This causes the AC to have a harder time trying to get rid of the moisture in the classroom.

Primary stakeholders include teachers and students who have a class period to attend at the buildings G and K. These stakeholders who have classes in these buildings most likely have walked past these kinds of ACs, which are dripping a lot of water within a day, encountering this problem every day that the High School has a school day. This basically means that the AC will continue to drip and possibly rust and cause erosion each day, causing the custodians and technicians to come by these ACs and try to clean it up. These stakeholders will also have to set up this device and ensure to take it down when not using the AC by the end of the day. They will also have to make sure that the device does not overfill and clog up.

However, secondary stakeholders who may be bothered by our solutions will be the administrators who will have to approve of these designs if they ever become finalized, G and K building teachers and students who attend there, and custodians who will have to clean the device or ensure that it isn't leaking. These stakeholders will be affected by our solution the most due to them either having to pass by the device, trying to avoid tripping or ensuring to not bother the device and accidentally breaking or damaging the solution while it is attached to the AC due to this area being popular walkway due to students coming from the lower campus of MHS such as: Portables, J building, JROTC, or even their athletics area.

PROBLEM JUSTIFICATION

Currently, in G103, there is a measurable issue with G103's AC's condensation problem that causes the AC to drip. This problem occurs primarily when the AC is turned on at a humid time of day when the G103 teachers are occupying the classroom. According to a study by Nick's Air Conditioning & Heating, high outdoor humidity can contribute to air conditioner water leaks due to its impact on the cooling process and the condensation within the AC unit. Air conditioners cool the air and dehumidify it by removing moisture. The size of the issue is significant because each day that the High School has a school day or a teacher's day, the AC is used, and therefore, a puddle forms after an hour of use when it's humid. According to a study from Weather Spark, there was a high spike in humidity in April 2025 and August- November. If you observe the space, you will notice that there is a 1-foot puddle under the AC. Causing erosion and possible mold growth underneath the AC unit. According to the EPA, mold reproduces by means of tiny airborne spores; the spores are invisible to the naked eye and float through outdoor and indoor air. This can harm the students who are near the area for a long amount of time, for example, during lunch, recess, or after school activities.

Sources:

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